

Research Methodology

CSE 505

Credit:2.0

Offered in Jan-Jun Semester

Course Content

Scope of survey-based research: Types of surveys – specific, periodic and transaction- driven, identification of research problem, analysis of research problem, customer identification, categorization and sampling. Planning of survey project – resources, budget and schedule. Preparation of questionnaire – elements of questionnaire, sequencing questions, and question formats. Methods of conducting survey, data collection, analysis and compilation of survey report.

Experimental and modeling skills: Introduction, selection of variables, design matrix, 2-level factorial design, 3-level factorial design, fractional factorial design, analysis of variance, latest trends in experimental designs.

We Will Learn/Discuss

- What is research?
- Steps of research
- **Types of research**
- **Research methodologies**
- How to write research report/article?
- How to present research work?
- Authors & reviewers
- How to review?
- Plagiarism

What is Research

- Research refers to a creative and systematic study to enrich the knowledge level in a specific field for better world and use it to derive new application(s).
- A careful study of a subject, especially in order to discover new facts or information about it [1].
- Research is defined as the creation of new knowledge and/or the use of existing knowledge in a new and creative way so as to generate new concepts, methodologies and understandings. This could include synthesis and analysis of previous research to the extent that it leads to new and creative outcomes [2].

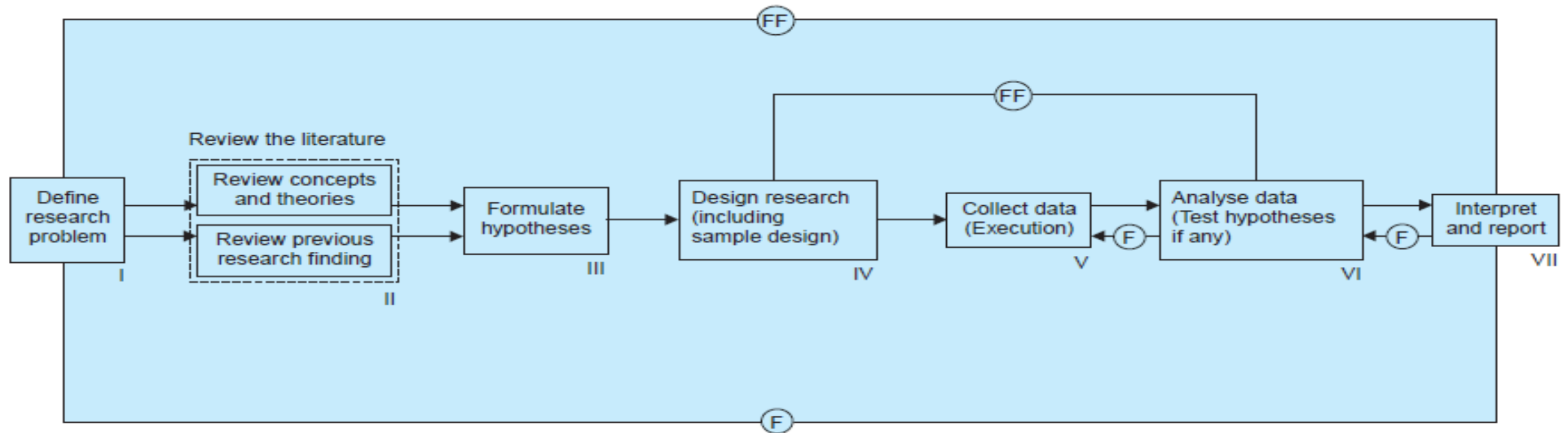
What is Research (contd.)

- In science, research is the diligent systematic enquiry into nature and society to validate and refine existing knowledge and to generate new knowledge [3].

And Many More

Steps of Research

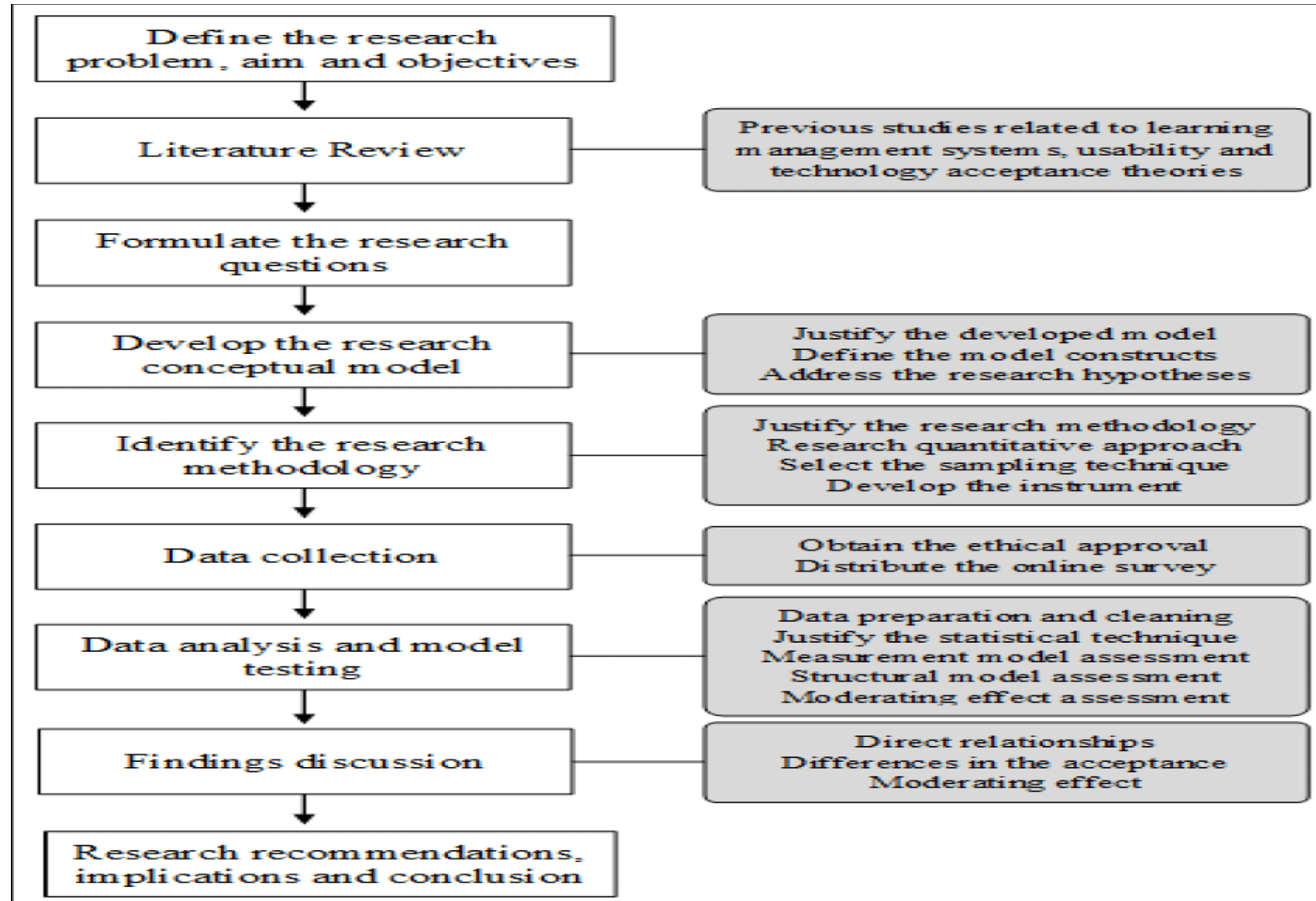
An extremely cyclic process



*Figure collected from the Internet

Where (F) = feed back (Helps in controlling the sub-system to which it is transmitted)
(FF) = feed forward (Serves the vital function of providing criteria for evaluation)

Steps of Research



Steps of Research (contd.)

- Motivation
- Research Objective
- Research Question
- Literature Review
- Decision of methodology
- Design/Development/Simulation
- Experiment/Analysis/Survey
- Result/Outcome/ Finding(s) analysis/interpret
- Write report/Submit to a conference/Presentation

Research Motivation

- The desire to solve unsolved problems.
- To find out what drives, wants, and needs people really have.
- To get intellectual joy of doing some creative work.
- The desire to serve the society.
- To discover new and test old scientific procedure as an efficient way to gain useful and fundamental knowledge.
- Curiosity about unknown things.
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Research Objective

- A clear and concise statement that defines the Goal/target/ aim/ purpose of a research.
- It describes concisely what the research is trying to achieve/accomplish.
- An idea/plan/purpose to understand deeply a subject matter by performing a research.
- Summary of research goal.
- Indicates what would be the expected outcome of a research.
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Research Question

- The core question that is aimed to answer throughout the research.
- Choosing a research question is an essential part of a research.
- Guides all the stages of a research.

❑ How does research objective differs from research question?

This is **NOT** called research

- ✓ gathering information from different sources such as books, journals, newspapers, magazines etc.
- ✓ No contribution to new knowledge.

Why to do research?
Who does research?

Reference List

1. https://www.oxfordlearnersdictionaries.com/definition/american_english/research_1
2. https://www.westernsydney.edu.au/research/researchers/preparing_a_grant_application/def_definition_of_research
3. Naidoo, Navindhra. (2011). What is research? A conceptual understanding. Journal of Emergency Medicine. 1. 10.1016/j.afjem.2011.05.011.